Quantum Gravitational Dynamics: The Formulation of Quantum Gravity from the Dirac Equation

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Abstract

We demonstrate that galaxy rotation curves across 175 SPARC galaxies (4,248 measurements) are reproduced with R^2=0.908 and zero free parameters per galaxy by a quantum correction to the gravitational potential derived from the WKB phase of the Dirac wavefunction

1 Introduction

The central question addressed in this paper is how classical gravity emerges from quantum mechanics. The proposed answer rests on a specific mathematical structure: the wavefunction encodes gravitational dynamics, and current conservation provides the bridge to classical physics.

The point of departure is the observation that gravity may be modelled as a spherical matter wave radiating to infinity. Any such wave in three spatial dimensions carries a $1/r$ amplitude factor from geometry alone. The wavefunction is therefore taken to have the form

$$\psi = \frac{\alpha_G}{r}\, u\, e^{iS/\hbar} \tag{1}$$

giving $|\psi|^2 \propto 1/r^2$. Combined with the current-conservation relation

$$|\psi|^2 \cdot p \cdot r^2 = C \tag{2}$$

the r^2 factors cancel, yielding a momentum equation determined by geometry alone, without a plane-wave postulate.

The analysis is carried out in *flat Minkowski spacetime*; no curved background geometry is assumed.

Note on the gravitational coupling constant. Throughout this paper the single symbol α_G denotes the gravitational fine structure constant. Its fundamental (complex) value, derived in Section 11 from phase-oscillation analysis, is

$$\boxed{\alpha_G = e^{i\pi/4}\sqrt{\frac{c\hbar}{2GMm}} = \frac{1+i}{\sqrt{2}}\sqrt{\frac{c\hbar}{2GMm}}} \tag{3}$$

so that

$$\alpha_G^2 = \frac{ic\hbar}{2GMm}, \qquad |\alpha_G|^2 = \frac{c\hbar}{2GMm}. \tag{4}$$

Because α_G is complex, it appears in the wavefunction and phase equations in its full form. Wherever a *real* quantity is required—probability densities, physical forces, normalization constants—one uses $|\alpha_G|^2$ or

$$\mathrm{Re}(\alpha_G) = \frac{1}{\sqrt{2}}\sqrt{\frac{c\hbar}{2GMm}} = \sqrt{\frac{c\hbar}{4GMm}}. \tag{5}$$

Dimensional check: $[\alpha_G^2] = [c\hbar/(GMm)] = 1$. ✓

2 Microscopic Foundation

2.1 Field Content and Dynamics

The theory is formulated in terms of a single microscopic degree of freedom: the Dirac spinor $\psi(x) \in \mathbb{C}^4$. The microscopic action in flat Minkowski spacetime is

$$\mathcal{S}_{\text{micro}} = \int d^4x \big[\bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi + G_{\text{NJL}}(\bar{\psi}\psi)^2 - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - e\bar{\psi}\gamma^\mu A_\mu\psi\big] \tag{6}$$

where $\bar{\psi} = \psi^\dagger\gamma^0$, $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ with $\eta = \text{diag}(-1, +1, +1, +1)$, and G_{NJL} is the Nambu–Jona-Lasinio (NJL) four-fermion coupling constant. The four terms are, respectively: the free Dirac kinetic term, a local four-fermion self-interaction, the Maxwell kinetic term, and the minimal electromagnetic coupling.

The free-field subsector

$$S_D = \int d^4x\, \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi \tag{7}$$

is the unique Lorentz-invariant, local, linear action for spin-$\frac{1}{2}$ fields with positive-definite energy, but it does not suffice as the microscopic starting point: coarse-graining S_D alone generates only mass and wavefunction renormalisation, since no new interaction operators can arise from a free theory. The pressure term $P(\bar{\psi}\psi)$, Maxwell kinetic term, and angular-momentum coupling $J^2\bar{\psi}\psi$ that appear in the gravitational effective Lagrangian all require the interacting terms present in $\mathcal{S}_{\text{micro}}$. Specifically, Wilsonian coarse-graining of $\mathcal{S}_{\text{micro}}$ generates $P(\bar{\psi}\psi)$ from the four-Fermi term, $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ from the EM sector, the J^2 operator from its derivative expansion, and ρ_Λ from zero-point fluctuations. All coarse-graining results in the subsequent sections are derived from $\mathcal{S}_{\text{micro}}$.

2.2 Stress-Energy Tensor

The symmetric stress-energy tensor:

$$T^{\mu\nu} = \frac{i}{4}\bar{\psi}\Big(\gamma^\mu \overleftrightarrow{\partial}^\nu + \gamma^\nu \overleftrightarrow{\partial}^\mu\Big)\psi \tag{8}$$

where $\overleftrightarrow{\partial} = \overrightarrow{\partial} - \overleftarrow{\partial}$. On-shell: $\partial_\mu T^{\mu\nu} = 0$.

This tensor provides the matter content that sources gravitational effects—even though we remain in flat spacetime.

3 Coarse-Graining and the Emergence of the Effective Theory

3.1 Scale Hierarchy

The theory possesses three characteristic length scales. Two of them are given at the outset:

- Λ_{UV}^{-1}: the ultraviolet length scale at which the effective description breaks down. Its physical identity is *not assumed here*; it will be derived in Section 24 as a consequence of the coarse-graining procedure and the structure of the loop integrals.

- L_{grav}: the scale over which gravitational gradients are appreciable, i.e. the scale on which $\nabla T^{\mu\nu}/T^{\mu\nu} \sim 1/L_{\text{grav}}$.

The coarse-graining scale ℓ is then any scale satisfying

$$\Lambda_{\text{UV}}^{-1} \ll \ell \ll L_{\text{grav}}. \tag{9}$$

This hierarchy is a *consistency requirement*, not an assumption: the Wilsonian procedure of Section 3.3 is self-consistent if and only if (9) holds. When $\ell \sim \Lambda_{\rm UV}^{-1}$, fast-mode fluctuations are not suppressed and the effective description fails; when $\ell \sim L_{\rm grav}$, the coarse-grained theory cannot resolve the gravitational gradients it is meant to describe.

The identification $\Lambda_{\rm UV}^{-1} = \hbar/(mc)$ is a *result*, not an input. It will emerge in Section 24 from the requirement that the effective mass generated by the loop integrals matches the physical mass m. We deliberately leave $\Lambda_{\rm UV}$ as a free UV scale throughout the present section to avoid any circularity.

3.2 Full Theory and Regularisation

The starting point for the Wilsonian coarse-graining is the path integral over $\mathcal{S}_{\rm micro}$ (6) in flat Minkowski spacetime $(\mathcal{M}, \eta_{\mu\nu})$, signature $(-,+,+,+)$:

$$\mathcal{Z} \;=\; \int \mathcal{D}\bar\psi\,\mathcal{D}\psi\;\exp\!\left(\frac{i}{\hbar}\int d^4x\,\left[\bar\psi\big(i\hbar\gamma^\mu\partial_\mu - mc\big)\psi + G_{\rm NJL}(\bar\psi\psi)^2 - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - e\bar\psi\gamma^\mu A_\mu\psi\right]\right), \tag{10}$$

with a UV cutoff $\Lambda_{\rm UV}$ imposed by Pauli–Villars regularisation. No assumption is made about the value or interpretation of $\Lambda_{\rm UV}$ at this stage.

3.3 Mode Decomposition and Integration over Fast Modes

3.3.1 Splitting the field

Introduce a sharp momentum cutoff at $\mu = 1/\ell$ and decompose:

$$\psi(\mathbf{x}) \;=\; \psi_<(\mathbf{x}) + \psi_>(\mathbf{x}), \tag{11}$$

where

$$\psi_<(\mathbf{x}) = \int_{|\mathbf{k}|<\mu} \frac{d^4k}{(2\pi)^4}\,\tilde\psi(\mathbf{k})\,e^{ik\cdot x}, \tag{12}$$

$$\psi_>(\mathbf{x}) = \int_{\mu<|\mathbf{k}|<\Lambda_{\rm UV}} \frac{d^4k}{(2\pi)^4}\,\tilde\psi(\mathbf{k})\,e^{ik\cdot x}. \tag{13}$$

The slow field $\psi_<$ encodes physics on scales $\geq \ell$; the fast field $\psi_>$ encodes physics between $\Lambda_{\rm UV}^{-1}$ and ℓ.

3.3.2 Gaussian integration over fast modes

The fast-mode integral is evaluated by separating the action into a quadratic (free Dirac) part and the interaction terms. The free part generates a Gaussian integral over $\psi_>$ that can be performed exactly; the NJL four-Fermi interaction and the EM coupling contribute via an expansion of $e^{iS_{\rm int}/\hbar}$ in powers of $G_{\rm NJL}$ and e respectively, generating the loop diagrams that produce the effective operators listed below. The result is

$$\mathcal{Z} \;=\; \int \mathcal{D}\bar\psi_<\,\mathcal{D}\psi_<\;\exp\!\left(\frac{i}{\hbar}S_{\rm eff}[\psi_<]\right), \tag{14}$$

where the effective action is

$$S_{\rm eff}[\psi_<] \;=\; S_D[\psi_<] \;-\; i\hbar\,{\rm Tr}\ln\!\big(i\hbar\gamma^\mu\partial_\mu - mc - V[\psi_<]\big)\Big|_>, \tag{15}$$

$|_>$ denotes restriction to the momentum shell $[\mu, \Lambda_{\rm UV}]$, and $V[\psi_<]$ encodes the interaction vertices of $\mathcal{S}_{\rm micro}$ with slow-mode external legs. The Tr ln generates all one-particle-irreducible (1PI) diagrams with fast-mode internal lines and slow-mode external legs.

Generation of effective operators Expanding S_{eff} in (15) in powers of $\psi_<$ and retaining operators of mass dimension ≤ 4 (marginal and relevant under the RG, cf. Section 3.4), subject to Lorentz invariance, U(1) phase symmetry, and locality, yields

$$\mathcal{L}_{\text{eff}} = \frac{i}{2}\bar{\psi}_<\gamma^\mu \overleftrightarrow{\partial}_\mu \psi_< - m_{\text{eff}}\,\bar{\psi}_<\psi_< - P(\bar{\psi}_<\psi_<) - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2M^2}J_{\mu\nu}J^{\mu\nu}\bar{\psi}_<\psi_< - \rho_\Lambda, \quad (16)$$

where $m_{\text{eff}} = m + \delta m(\Lambda_{\text{UV}}, \mu)$ contains the mass renormalisation from fast-mode loops, with δm depending explicitly on Λ_{UV}.

The loop expansion of (15) generates the following operators, each traceable to a specific diagram topology:

1. **Mass renormalisation.** The one-loop self-energy with two external slow legs shifts

$$\delta m = \frac{3e^2}{16\pi^2}\int_\mu^{\Lambda_{\text{UV}}} \frac{d^4k}{(2\pi)^4}\frac{1}{k^2 - m^2c^2} \sim \frac{3e^2\Lambda_{\text{UV}}^2}{16\pi^2 mc^2}. \quad (17)$$

   The physical mass is defined by the renormalisation condition $m_{\text{eff}}|_{\mu=0} = m_{\text{phys}}$, which fixes the counterterm and eliminates δm from the final theory. The scale at which δm becomes comparable to m will identify Λ_{UV} (see Section 24).

2. **Pressure.** The four-Fermi vertex $G_{\text{NJL}}(\bar{\psi}\psi)^2$ present in $\mathcal{S}_{\text{micro}}$ generates, via a four-external-leg slow-mode diagram with one fast internal loop, the term $-g_P(\bar{\psi}_<\psi_<)^2$, which at low energy becomes the equation-of-state term $P(\bar{\psi}_<\psi_<)$.

3. **Electromagnetic coupling.** The minimal coupling $-e\bar{\psi}\gamma^\mu A_\mu\psi$ present in $\mathcal{S}_{\text{micro}}$ is the unique U(1)-covariant marginal deformation; integrating out fast charged modes generates the Maxwell kinetic term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ in the effective theory.

4. **Angular momentum / spin.** The dimension-6 operator $J_{\mu\nu}J^{\mu\nu}\bar{\psi}_<\psi_</(2M^2)$ arises from box diagrams; it encodes frame-dragging effects in the stress-energy tensor.

5. **Vacuum energy.** Zero-point fluctuations of fast modes contribute $\rho_\Lambda = \hbar\int_\mu^{\Lambda_{\text{UV}}} d^4k/(2\pi)^4 \cdot \omega_{\mathbf{k}}$, a constant shift whose IR value is set by matching to the observed cosmological constant.

6. **No higher operators.** All operators of mass dimension > 4 carry factors of $(\ell\,\Lambda_{\text{UV}})^{-(d-4)} \ll 1$ under the hierarchy (9) and are therefore suppressed. ■

3.4 Wilsonian Renormalisation Group Flow

3.4.1 RG transformation

Define the Wilsonian RG step as: (i) integrate out the momentum shell $|\mathbf{k}| \in [b^{-1}\mu, \mu]$, $b > 1$; (ii) rescale

$$x \to bx, \qquad \psi_< \to b^{-3/2}\psi_<, \quad (18)$$

to restore the kinetic term to canonical normalisation. Let $t = \ln b$.

3.4.2 Beta functions

The one-loop RG equations for the couplings in (16) are:

$$\frac{dm}{dt} = m\,(1+\gamma_m)\,, \qquad \gamma_m = -\frac{3e^2}{16\pi^2}, \tag{19}$$

$$\frac{dg_P}{dt} = g_P(1-2\gamma_\psi) + \frac{N_f g_P^2}{8\pi^2}, \tag{20}$$

$$\frac{de^2}{dt} = \frac{e^4}{6\pi^2}, \tag{21}$$

$$\frac{d(1/M^2)}{dt} = -\frac{2}{M^2}\,(1-\gamma_J)\,, \tag{22}$$

$$\frac{d\rho_\Lambda}{dt} = 4\rho_\Lambda + \frac{N_f\,\mu^4}{16\pi^2}. \tag{23}$$

Here N_f is the number of fermion flavours and γ_ψ, γ_J are the anomalous dimensions of $\bar\psi\psi$ and $J_{\mu\nu}$ respectively.

IR fixed point and decoupling In the limit $t \to \infty$ (i.e. $\ell \to L_{\text{grav}}$):

1. The dimension-6 coupling $1/M^2 \to 0$: angular-momentum corrections are suppressed as $(\ell/L_{\text{grav}})^2$.
2. ρ_Λ reaches its IR value fixed by cosmological matching.
3. Quantum fluctuations average to zero: $\langle\psi_>\rangle = 0$, leaving classical effective fields.
4. The physical mass m_{phys} is stable, with δm absorbed by the renormalisation condition.

The coupling $1/M^2$ has canonical dimension -2; under rescaling (18) it acquires a factor b^{-2}, so it is power-law irrelevant and flows to zero. The remaining claims follow from the Appelquist–Carazzone decoupling theorem: fields with masses $\gg \mu$ decouple from low-energy physics, ensuring $\langle\psi_>\rangle = 0$ at scales $\ell \gg \Lambda_{\text{UV}}^{-1}$. ■

3.5 The Effective Lagrangian

Effective Lagrangian from Wilsonian coarse-graining Let the interacting Dirac theory $\mathcal{S}_{\text{micro}}$ (6) be coarse-grained over a scale ℓ satisfying (9), with Λ_{UV} an unspecified UV scale. Then the unique effective Lagrangian consistent with Lorentz invariance, U(1) symmetry, locality, and positive-definite energy, retaining only marginal and relevant operators, is

$$\boxed{\mathcal{L}_{\text{eff}} \;=\; \frac{i}{2}\bar\psi\gamma^\mu\overleftrightarrow{\partial}_\mu\psi \;-\; m\bar\psi\psi \;-\; P(\bar\psi\psi) \;-\; \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \;-\; \frac{1}{2M^2}J_{\mu\nu}J^{\mu\nu}\bar\psi\psi \;-\; \rho_\Lambda} \tag{24}$$

(slow-mode subscripts dropped). Each term has a precise gravitational interpretation:

Term	**Gravitational limit**
$m\bar\psi\psi$	Rest-mass density $\to$ Schwarzschild
$P(\bar\psi\psi)$	Pressure $\to$ equation of state
$\frac{1}{2M^2}J_{\mu\nu}J^{\mu\nu}\bar\psi\psi$	Angular momentum $\to$ Kerr frame-dragging
$\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$	EM energy $\to$ Reissner–Nordström
ρ_Λ	Vacuum energy $\to$ cosmological constant

This correspondence is a *result* of the coarse-graining, not a postulate.

Proposition 3.3.2 establishes generation of all operators in (24). Proposition 3.4.2 establishes that no further operators survive at IR scales. Uniqueness follows from the exhaustive classification of Lorentz-invariant, U(1)-symmetric, local bilinears and quadratics in ψ, $F_{\mu\nu}$, $J_{\mu\nu}$ of mass dimension ≤ 4: every entry in (24) appears in this list, and the list is finite. Higher-dimension operators are suppressed by $(\Lambda_{\text{UV}}^{-1}/\ell)^{d-4} \ll 1$ under (9). ■

3.6 Connection to the Phase Gradient and Newton's Law

After coarse-graining, apply the WKB ansatz to the slow mode:

$$\psi_<(\mathbf{x}) \;=\; R(\mathbf{x})\, e^{iS(\mathbf{x})/\hbar} \tag{25}$$

where R varies on scales $\sim L_{\text{grav}}$ and S oscillates on scales $\gg \ell$. The phase gradient σ_μ is the effective classical variable from which the metric is reconstructed.

Applying Noether's theorem to $\mathcal{L}_{\text{eff}}$ and averaging over ℓ yields the coarse-grained stress-energy tensor:

$$\boxed{\langle T^{\mu\nu}\rangle \;=\; \rho c^2 u^\mu u^\nu \;+\; P g^{\mu\nu} \;+\; \langle \tau^{\mu\nu}_{\text{spin}}\rangle \;+\; \langle \tau^{\mu\nu}_{\text{EM}}\rangle \;+\; \rho_\Lambda g^{\mu\nu}.} \tag{26}$$

The result (26) has the structure of the Einstein stress-energy tensor; each term arises from the corresponding term in the effective Lagrangian via coarse-graining rather than being postulated independently.

The gravitational force function then takes the form $F(r) = \Omega/P(r)$ where $P(r) = i\sinh(2\alpha r)$ emerges from the phase-gradient equation; Newton's law is the $\alpha r \to 0$ limit (see Section 11 and Section 24 where α — and hence Λ_{UV}^{-1} — is identified).

3.7 Effective Lagrangian

The effective Lagrangian derived in Theorem 3.5, Eq. (24), is the unique result consistent with Lorentz invariance, U(1) symmetry, locality, and positive-definite energy. To summarise its physical content: each term maps to a specific classical gravitational effect in the macroscopic limit—rest-mass density to Schwarzschild, pressure to equation of state, angular momentum to Kerr frame-dragging, electromagnetic energy to Reissner–Nordström, and vacuum energy to the cosmological constant. The derivation and full table are given in Theorem 3.5.

4 The Complete Action

$$\boxed{S = \int d^4x \left[\frac{i}{2}\bar\psi\gamma^\mu \overleftrightarrow{\partial}_\mu \psi - m\bar\psi\psi - P(\bar\psi\psi) - \frac{1}{2M^2} J_{\mu\nu}J^{\mu\nu}\bar\psi\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \rho_\Lambda\right]} \tag{27}$$

where:

- $\bar\psi = \psi^\dagger\gamma^0, \quad \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$
- $\overleftrightarrow{\partial}_\mu = \overrightarrow{\partial}_\mu - \overleftarrow{\partial}_\mu$
- $J_{\mu\nu} = x_\mu p_\nu - x_\nu p_\mu$ (angular momentum tensor)
- $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ (electromagnetic field tensor)

5 Stress-Energy Tensor from Action

5.1 Noether's Theorem

From spacetime translation invariance:

$$T^{\mu\nu} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\partial^\nu\psi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\bar\psi)}\partial^\nu\bar\psi - g^{\mu\nu}\mathcal{L} \tag{28}$$

After symmetrization and coarse-graining over scale ℓ:

$$\boxed{\langle T^{\mu\nu}\rangle = \rho(x)c^2u^\mu u^\nu + P(x)g^{\mu\nu} + \langle\tau^{\mu\nu}_{\text{spin}}\rangle + \langle\tau^{\mu\nu}_{\text{EM}}\rangle + \rho_\Lambda g^{\mu\nu}} \tag{29}$$

The result has the structure of the Einstein stress-energy tensor.

5.2 Component Origins

Each stress-energy term traces to an action term:

$$\rho c^2 u^\mu u^\nu \leftarrow \tfrac{i}{2}\bar{\psi}\gamma^\mu \overleftrightarrow{\partial}_\mu \psi - m\bar{\psi}\psi \tag{30}$$
$$P g^{\mu\nu} \leftarrow -P(\bar{\psi}\psi) \tag{31}$$
$$\tau^{\mu\nu}_{\text{spin}} \leftarrow -\tfrac{1}{2M^2} J_{\mu\nu} J^{\mu\nu} \bar{\psi}\psi \tag{32}$$
$$\tau^{\mu\nu}_{\text{EM}} \leftarrow -\tfrac{1}{4} F_{\mu\nu} F^{\mu\nu} \tag{33}$$
$$\rho_\Lambda g^{\mu\nu} \leftarrow -\rho_\Lambda \tag{34}$$

6 Modified Dirac Equation and Energy Denominators

6.1 Equation of Motion

Varying the action with respect to $\bar{\psi}$:

$$i\gamma^\mu \partial_\mu \psi = \left[m + \frac{\partial P}{\partial(\bar{\psi}\psi)} + \frac{1}{2M^2} J_{\mu\nu} J^{\mu\nu} \right] \psi \tag{35}$$

For stationary states $\psi = u(p) e^{-iS/\hbar}$:

$$[\gamma^\mu p_\mu - mc\gamma^0]\psi = V_{\text{eff}} \gamma^0 \psi \tag{36}$$

6.2 The Effective Hamiltonian

All stress-energy contributions enter through:

$$\boxed{H = \int \rho c^2 \, dV - \int P \, dV - \frac{J^2}{2mr^2} + V_{EM} + \rho_\Lambda} \tag{37}$$

where each term originates from the effective Lagrangian:

- $\int \rho c^2 \, dV$: from $\frac{i}{2}\bar{\psi}\gamma^\mu \overleftrightarrow{\partial}_\mu \psi$ (source mass-energy)
- $-\int P \, dV$: from $-P(\bar{\psi}\psi)$ (pressure work)
- $-J^2/(2mr^2)$: from $-\frac{1}{2M^2} J_{\mu\nu} J^{\mu\nu} \bar{\psi}\psi$ (frame-dragging)
- V_{EM}: from $-\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ (electromagnetic potential)
- ρ_Λ: from $-\rho_\Lambda$ (vacuum energy)

6.3 Energy Denominators

This defines the energy denominators:

$$\Delta = E + mc^2 + H \quad \text{(particle states)} \tag{38}$$
$$\Delta' = E - mc^2 - H \quad \text{(antiparticle states)} \tag{39}$$

Explicit form with all contributions:

$$\boxed{\Delta = E + \underbrace{mc^2}_{\text{test}} + \underbrace{\int \rho c^2 \, dV}_{\text{source}} - \underbrace{\int P \, dV}_{\text{pressure}} - \underbrace{\frac{J^2}{2mr^2}}_{\text{spin}} + \underbrace{V_{EM}}_{\text{EM}} + \underbrace{\rho_\Lambda}_{\text{vacuum}}} \tag{40}$$

7 The Quantum-Classical Bridge

7.1 Macroscopic Coherent States and the Spherical Wave Ansatz

Gravity is treated as a **spherical matter wave radiating to infinity**. Any wave radiating from a point source in three spatial dimensions must spread over a growing spherical surface of area $4\pi r^2$. Flux conservation therefore demands that the wavefunction amplitude falls as $1/r$. A macroscopically coherent gravitational state must take the form:

$$\psi = \frac{\alpha_G}{r}\, e^{iS(x)/\hbar}\, u \tag{41}$$

where:

- α_G/r: amplitude — the $1/r$ is **geometric**, mandatory for any spherical wave in 3D
- $S(x)$: rapidly-varying phase (the action)
- u: slowly-varying four-component spinor envelope

Dimensional note. Previously the wavefunction was written $\psi = \alpha_G \cdot u \cdot e^{iS/\hbar}$ with α_G carrying no r-dependence. This was the source of the earlier inconsistency: α_G was dimensionless but ψ requires dimensions $L^{-3/2}$, so the spatial structure was implicitly hidden in u. Explicitly writing $1/r$ restores dimensional consistency and follows from the spherical-wave geometry.

7.2 Probability Current

The Dirac current with the spherical wave ansatz:

$$j^\mu = \bar{\psi}\gamma^\mu\psi = \frac{|\alpha_G|^2}{r^2}\,\bar{u}\gamma^\mu u \tag{42}$$

Note: The phase $e^{iS/\hbar}$ cancels in $\bar{\psi}\psi = \psi^\dagger\gamma^0\psi$.
Non-relativistic limit: For spinor aligned with momentum:

$$j^0 \approx \frac{|\alpha_G|^2}{r^2}, \qquad \mathbf{j} \approx \frac{|\alpha_G|^2}{r^2}\frac{\nabla S}{m} \tag{43}$$

The total outward probability flux through any sphere of radius r is:

$$\Phi = 4\pi r^2 \cdot j^r = 4\pi|\alpha_G|^2 \cdot p = \text{const} \tag{44}$$

confirming that $|\alpha_G|^2 \sim$ kg/s is a conserved flux, as required.

7.3 Current Conservation: The Central Equation

From $\partial_\mu j^\mu = 0$, for stationary states:

$$\boxed{\nabla \cdot \big(|\psi|^2 \nabla S\big) = 0} \tag{45}$$

7.4 Spherical Symmetry: The Exact Bridge

For spherically symmetric configurations with $\nabla S = p(r)\hat{r}$:

$$\frac{1}{r^2}\frac{\partial}{\partial r}\big(r^2|\psi|^2 p(r)\big) = 0 \tag{46}$$

Integrating gives the **exact** conservation law:

$$\boxed{|\psi|^2 \cdot p(r) \cdot r^2 = C} \tag{47}$$

Substituting $|\psi|^2 = (|\alpha_G|^2/r^2)(1 + p^2c^2/\Delta^2)$ from the spherical wave ansatz:

$$\frac{|\alpha_G|^2}{r^2}\left(1 + \frac{p^2}{\Delta^2}\right) \cdot p \cdot r^2 = C \tag{48}$$

The r^2 cancels exactly from geometry:

$$\boxed{|\alpha_G|^2\left(p + \frac{p^3}{\Delta^2}\right) = C} \tag{49}$$

Equation (49) is the quantum-classical bridge. The r^2 cancellation follows from the spherical-wave ansatz and is exact; no plane-wave approximation is introduced. The same geometric factor that produces Newton's $1/r^2$ law is responsible for the cancellation.

The constant C will be determined through the complete system of equations.

Remark 7.1 (The bridge holds for all r; the closing equation changes with regime). The current-conservation bridge $|\psi|^2 \cdot p \cdot r^2 = C$ is *exact* at every length scale—it follows from $\partial_\mu j^\mu = 0$ alone and requires no approximation. It provides one equation in two unknowns $\{p(r), |\psi|(r)\}$ and must be paired with a second, dynamical equation. That second equation changes with the physical regime:

Regime	**Dynamical equation**	**Result**
$r \sim \lambda_C$ (quantum gravity)	Phase expansion of $e^{2ipr/\hbar}$ + gravitational matching	Newton's law *and* quantum corrections $F \propto (\lambda_C/r)^{2n}$
$r \gg \lambda_C$, classical	QGD field equation $\Box_g\sigma = \text{source}$, solution $\sigma_t = \sqrt{r_s/r}$	Newton's law (0PN), exact for all macroscopic r
$r \gg \lambda_C$, relativistic	QGD propagator loop integrals $\Delta_{\text{QGD}}(k) = 1/k^2 - 1/(k^2 + m_Q^2)$	Full PN potential $A(u;\nu)$ to all orders

The bridge $|\psi|^2 \cdot p \cdot r^2 = C$ is therefore the **universal skeleton** spanning all three regimes and all distance scales. Demonstrating this unified coverage explicitly is a goal of the ongoing QGD programme.

8 Complete Four-Component Wavefunction Solution

8.1 Eigenvalue Problem with Full Stress-Energy

In Pauli-Dirac representation $u = (u_A, u_B)^T$:

$$\begin{pmatrix} E - mc^2 - H & -c\boldsymbol{\sigma}\cdot\mathbf{p} \\ c\boldsymbol{\sigma}\cdot\mathbf{p} & -(E + mc^2 + H) \end{pmatrix}\begin{pmatrix} u_A \\ u_B \end{pmatrix} = 0 \tag{50}$$

where H contains all stress-energy contributions as defined in Eq. (37).

8.2 General Solution with All Stress-Energy Terms

The most general solution, with α_G as the universal prefactor, is:

$$\psi = \frac{\alpha_G}{r}\left[\psi_0 \begin{bmatrix} 1 \\ 0 \\ \sigma_z \\ \sigma_x + i\sigma_y \end{bmatrix} e^{-iS/\hbar} + \psi_1 \begin{bmatrix} 0 \\ 1 \\ \sigma_x - i\sigma_y \\ -\sigma_z \end{bmatrix} e^{-iS/\hbar} + \psi_2 \begin{bmatrix} \sigma'_z \\ \sigma'_x + i\sigma'_y \\ 1 \\ 0 \end{bmatrix} e^{+iS/\hbar} + \psi_3 \begin{bmatrix} \sigma'_x - i\sigma'_y \\ -\sigma'_z \\ 0 \\ 1 \end{bmatrix} e^{+iS/\hbar}\right] \tag{51}$$

where the **graviton scalars** with minimal electromagnetic coupling are:

$$\sigma_i = \frac{(p_i - eA_i/c)c}{\Delta} \quad \text{(particle states)} \tag{52}$$

$$\sigma'_i = \frac{(p_i - eA_i/c)c}{\Delta'} \quad \text{(antiparticle states)} \tag{53}$$

and Δ, Δ' contain all stress-energy contributions from Eq. (40).

Note on reality: α_G is complex; in all probability-density computations we use $|\alpha_G|^2 = c\hbar/(2GMm)$, not α_G^2.

8.3 Unified Form (Single-Component, Particle-Only)

$$\boxed{\psi = \frac{\alpha_G}{r}\begin{bmatrix} 1 \\ 0 \\ \dfrac{(p_z - eA_z/c)c}{\Delta_{\text{full}}} \\ \dfrac{[(p_x - eA_x/c) + i(p_y - eA_y/c)]c}{\Delta_{\text{full}}} \end{bmatrix} e^{-iS/\hbar}} \tag{54}$$

where:

$$\boxed{\Delta_{\text{full}} = E + mc^2 + \int \rho c^2\, dV - \int P\, dV - \frac{J^2}{2mr^2} + V_{EM} + \rho_\Lambda} \tag{55}$$

All five stress-energy contributions (ρ, P, J, F, Λ) enter through the same denominator Δ_{full}.

9 Special Cases: Classical Limits

9.1 Pure Schwarzschild ($P = J = F = \Lambda = 0$)

$$\Delta_{\text{Schw}} = E + mc^2 + \int \rho c^2\, dV \tag{56}$$

$$\psi_{\text{Schw}} = \frac{\alpha_G}{r}\begin{bmatrix} 1 \\ 0 \\ p_z c/\Delta_{\text{Schw}} \\ (p_x + ip_y)c/\Delta_{\text{Schw}} \end{bmatrix} e^{-iS/\hbar} \tag{57}$$

Stress-energy: $T^{\mu\nu} = \rho c^2 u^\mu u^\nu$

9.2 Kerr: Rotating Black Hole ($P = F = \Lambda = 0$, $J \neq 0$)

$$\Delta_{\text{Kerr}} = E + mc^2 + \int \rho c^2\, dV - \frac{J^2}{2mr^2} \tag{58}$$

$$\psi_{\text{Kerr}} = \frac{\alpha_G}{r}\begin{bmatrix} 1 \\ 0 \\ p_z c/\Delta_{\text{Kerr}} \\ (p_x + ip_y)c/\Delta_{\text{Kerr}} \end{bmatrix} e^{-iS/\hbar} \tag{59}$$

Stress-energy: $T^{\mu\nu} = \rho c^2 u^\mu u^\nu + \tau^{\mu\nu}_{\text{spin}}$

9.3 Reissner-Nordström: Charged Black Hole ($P = J = \Lambda = 0$, $F \neq 0$)

$$\Delta_{\text{RN}} = E + mc^2 + \int \rho c^2 \, dV + V_{EM} \tag{60}$$

$$\psi_{\text{RN}} = \frac{\alpha_G}{r} \begin{bmatrix} 1 \\ 0 \\ (p_z - eA_z/c)c/\Delta_{\text{RN}} \\ [(p_x - eA_x/c) + i(p_y - eA_y/c)]c/\Delta_{\text{RN}} \end{bmatrix} e^{-iS/\hbar} \tag{61}$$

Stress-energy: $T^{\mu\nu} = \rho c^2 u^\mu u^\nu + \tau^{\mu\nu}_{\text{EM}}$

9.4 Perfect Fluid with Cosmological Constant ($J = F = 0$)

$$\Delta_{\text{fluid}} = E + mc^2 + \int \rho c^2 \, dV - \int P \, dV + \rho_\Lambda \tag{62}$$

$$\psi_{\text{fluid}} = \frac{\alpha_G}{r} \begin{bmatrix} 1 \\ 0 \\ p_z c/\Delta_{\text{fluid}} \\ (p_x + ip_y)c/\Delta_{\text{fluid}} \end{bmatrix} e^{-iS/\hbar} \tag{63}$$

Stress-energy: $T^{\mu\nu} = \rho c^2 u^\mu u^\nu + P g^{\mu\nu} + \rho_\Lambda g^{\mu\nu}$

9.5 Most General Case

$$\boxed{\psi_{\text{general}} = \frac{\alpha_G}{r} \begin{bmatrix} 1 \\ 0 \\ \dfrac{(p_z - eA_z/c)c}{E + mc^2 + \int \rho c^2 \, dV - \int P \, dV - \frac{J^2}{2mr^2} + V_{EM} + \rho_\Lambda} \\ \dfrac{[(p_x - eA_x/c) + i(p_y - eA_y/c)]c}{E + mc^2 + \int \rho c^2 \, dV - \int P \, dV - \frac{J^2}{2mr^2} + V_{EM} + \rho_\Lambda} \end{bmatrix} e^{-iS/\hbar}} \tag{64}$$

10 Connecting Quantum and Classical

10.1 Probability Density

With the spherical wave ansatz $\psi = (\alpha_G/r)\, u\, e^{-iS/\hbar}$:

$$|\psi|^2 = \psi^\dagger \psi = \frac{|\alpha_G|^2}{r^2} \begin{bmatrix} 1 & 0 & \frac{p_z c}{\Delta} & \frac{(p_x + ip_y)c}{\Delta} \end{bmatrix}^\dagger \begin{bmatrix} 1 \\ 0 \\ \frac{p_z c}{\Delta} \\ \frac{(p_x + ip_y)c}{\Delta} \end{bmatrix} \tag{65}$$

$$= \frac{|\alpha_G|^2}{r^2} \left(1 + \frac{|p|^2 c^2}{\Delta^2} \right) \tag{66}$$

Therefore:

$$\boxed{|\psi|^2 = \frac{|\alpha_G|^2}{r^2} \left(1 + \frac{|p|^2 c^2}{\Delta^2} \right)} \tag{67}$$

where Δ contains all stress-energy contributions.

10.2 Applying the Exact Bridge

The exact conservation law (Eq. (47)) states $|\psi|^2 \cdot p \cdot r^2 = C$.

Substituting Eq. (67):

$$\frac{|\alpha_G|^2}{r^2}\left(1 + \frac{|p|^2 c^2}{\Delta^2}\right) \cdot |p| \cdot r^2 = C \tag{68}$$

The r^2 cancels exactly — a geometric consequence of the spherical wave:

$$|\alpha_G|^2 |p| + |\alpha_G|^2 \frac{c^2 |p|^3}{\Delta^2} = C \tag{69}$$

$$\boxed{|p| + \frac{c^2 |p|^3}{\Delta^2} = \frac{C}{|\alpha_G|^2}} \tag{70}$$

This cubic equation determines the momentum structure from the wavefunction. The cancellation is exact, not an approximation.

10.3 Non-Relativistic Limit

For $|p| \ll mc$, $\Delta \approx mc^2$, the cubic term is negligible:

$$\boxed{|p| \approx \frac{C}{|\alpha_G|^2}} \tag{71}$$

This connects the normalization constants to momentum.

11 Derivation of Newton's Law and Determination of α_G

11.1 Force from Wavefunction Phase Structure

The momentum and energy fields in the weak-field limit are:

$$p(r) = \frac{p_0}{\alpha_G^2} e^{-2ipr/\hbar}, \qquad E(r) = \frac{p_0 c}{\alpha_G^2 e^{2ipr/\hbar}}. \tag{72}$$

11.2 Phase Expansion

Taylor-expanding with $p_0 = mc$:

$$e^{2imcr/\hbar} = 1 + \frac{2imcr}{\hbar} - \frac{2m^2c^2r^2}{\hbar^2} - i\frac{4m^3c^3r^3}{3\hbar^3} + \cdots \tag{73}$$

11.3 Leading-Term Analysis

Using the first non-trivial term:

$$E(r) \approx \frac{mc^2}{\alpha_G^2 \cdot 2imcr/\hbar} = \frac{c\hbar}{2i\,\alpha_G^2\, r}. \tag{74}$$

Key step: this is where the free-Dirac theory couples to gravity

Up to this point α_G is an *undetermined* normalization constant of the free wavefunction. It becomes the gravitational coupling constant only when we **match the phase-field**

energy to the Newtonian gravitational potential of a source of mass M:

$$\underbrace{\frac{c\hbar}{2i\,\alpha_G^2\,r}}_{\text{phase-field energy (free QM)}} = \underbrace{-\frac{GMm}{r}}_{\text{gravitational potential (input)}} . \tag{75}$$

This matching is the *gravitational coupling step*: without it, α_G remains a free parameter and no gravitational physics is predicted. After this step, G, M, and m enter α_G and all subsequent quantities.

Note on the logical status of this step. The Newtonian potential $V = -GMm/r$ on the right-hand side is an *external physical input*, not derived from within QGD at this stage. It introduces Newton's constant G, the source mass M, and the test mass m as free parameters fixed by experiment. What QGD *does* derive from this matching is:

1. The specific form $\alpha_G^2 = ic\hbar/(2GMm)$, which fixes the wavefunction normalization in terms of the gravitational masses.
2. The $1/r^2$ falloff of the force (Section 11.11.5), which follows from $F = -dE/dr$ applied to $E \propto 1/r$ — a consequence of the spherical-wave geometry, not an additional input.
3. All quantum corrections beyond Newton (Section 24), which are genuine predictions with no classical analogue.

The recovery of $F = GMm/r^2$ in Section 11.11.5 therefore demonstrates *self-consistency* of QGD with Newton's law, not an independent derivation of it. The fundamental assumption is that classical Newtonian gravity holds at macroscopic scales; QGD provides the quantum-mechanical framework in which this is realised and extended.

For a static gravitational source, $E_{\text{field}} = V = -GMm/r$:

$$\frac{c\hbar}{2i\,\alpha_G^2\,r} = -\frac{GMm}{r}. \tag{76}$$

Solving:

$$\boxed{\alpha_G^2 = \frac{ic\hbar}{2GMm}} \tag{77}$$

11.4 The Complex Gravitational Coupling

Taking the square root using $\sqrt{i} = e^{i\pi/4} = (1+i)/\sqrt{2}$:

$$\boxed{\alpha_G = e^{i\pi/4}\sqrt{\frac{c\hbar}{2GMm}} = \frac{1+i}{\sqrt{2}}\sqrt{\frac{c\hbar}{2GMm}}} \tag{78}$$

Verification:

$$\alpha_G^2 = \frac{(1+i)^2}{2}\cdot\frac{c\hbar}{2GMm} = \frac{2i}{2}\cdot\frac{c\hbar}{2GMm} = \frac{ic\hbar}{2GMm} \quad \checkmark \tag{79}$$

Key derived quantities:

$$|\alpha_G|^2 = \frac{c\hbar}{2GMm} \tag{80}$$

$$\mathrm{Re}(\alpha_G) = \sqrt{\frac{c\hbar}{4GMm}} \tag{81}$$

11.5 Newton's Force from the Leading Term

$$F_1 = -\frac{dE}{dr} = \frac{c\hbar}{2i\,\alpha_G^2\,r^2}. \tag{82}$$

Substituting $\alpha_G^2 = ic\hbar/(2GMm)$:

$$F_1 = \frac{c\hbar}{2ir^2} \cdot \frac{2GMm}{ic\hbar} = \frac{GMm}{r^2} \quad \checkmark \tag{83}$$

Newton's inverse-square law is recovered exactly.

12 Determination of the Normalization Constant C

12.1 Classical Momentum at the Gravitational Bohr Radius

For a particle with zero total energy in a Newtonian potential:

$$p(r) = \sqrt{\frac{2GMm^2}{r}}. \tag{84}$$

The gravitational Bohr radius $a_0 = \hbar^2/(GMm^2)$ gives:

$$p(a_0) = \sqrt{\frac{2GMm^2 \cdot GMm^2}{\hbar^2}} = \frac{\sqrt{2}\,GMm^2}{\hbar}. \tag{85}$$

12.2 Value of C

From Eq. (71), $C = |\alpha_G|^2 \cdot p(a_0)$:

$$C = \frac{c\hbar}{2GMm} \cdot \frac{\sqrt{2}\,GMm^2}{\hbar} = \frac{\sqrt{2}\,cm \cdot m}{2m} = \frac{mc}{\sqrt{2}}. \tag{86}$$

$$\boxed{C = \frac{mc}{\sqrt{2}}} \tag{87}$$

Remarks: C has dimensions of momentum ✓. The flux scale is set purely by the test particle's rest momentum; source-mass dependence enters through α_G and Δ.

12.3 The Cubic Momentum Equation

Substituting C and $|\alpha_G|^2$ into Eq. (70):

$$\frac{C}{|\alpha_G|^2} = \frac{mc/\sqrt{2}}{c\hbar/(2GMm)} = \frac{mc}{\sqrt{2}} \cdot \frac{2GMm}{c\hbar} = \frac{\sqrt{2}\,GMm^2}{\hbar} \equiv \mathcal{P}. \tag{88}$$

$$\boxed{|p| + \frac{c^2|p|^3}{\Delta^2} = \mathcal{P} = \frac{\sqrt{2}\,GMm^2}{\hbar}} \tag{89}$$

13 Complete Gravitational Wavefunction

13.1 Normalized Form

$$\boxed{\psi(x) = \frac{\alpha_G}{r} \begin{bmatrix} 1 \\ 0 \\ p_z c/\Delta \\ (p_x + ip_y)c/\Delta \end{bmatrix} e^{-iS(r)/\hbar}} \tag{90}$$

where $\alpha_G = e^{i\pi/4}\sqrt{c\hbar/(2GMm)}$,

$$\Delta = E + mc^2 + \int \rho c^2 \, dV - \int P \, dV - \frac{J^2}{2mr^2} + V_{EM} + \rho_\Lambda, \tag{91}$$

and $|p|$ satisfies Eq. (89).

13.2 Hamilton-Jacobi Connection

The phase $S(r)$ satisfies:

$$\frac{1}{2m}\left(\frac{dS}{dr}\right)^2 + V(r) = E \tag{92}$$

with $V(r) = -GMm/r$. With $p(r) = dS/dr$:

$$F(r) = -\frac{dV}{dr} = -\frac{GMm}{r^2}. \tag{93}$$

This recovers Newton's law of gravitation.

14 Relation to Gravitational Potential

For completeness, with $\Phi(r) = -GMm/r$:

14.1 Coupling Constant in Terms of Potential

$$|\alpha_G|^2 = \frac{c\hbar}{2GMm} = -\frac{c\hbar}{2r\Phi(r)} \tag{94}$$

14.2 Wavefunction Amplitude

From the exact bridge $|\psi|^2 \cdot p \cdot r^2 = C$ with $p = \sqrt{-2m\Phi(r)}$ ($\Phi < 0$) and $|\psi|^2 = |\alpha_G|^2(1 + p^2c^2/\Delta^2)/r^2$:

$$|\psi|^2 \approx \frac{C}{r^2\,|p|} = \frac{mc/\sqrt{2}}{r^2\sqrt{2m|\Phi(r)|}} = \frac{c\sqrt{m}}{2r^2\sqrt{|\Phi(r)|}}. \tag{95}$$

$$\boxed{|\psi|^2 = \frac{c\sqrt{m}}{2\,r^2\sqrt{|\Phi(r)|}}} \tag{96}$$

Or in proportional form:

$$\boxed{|\psi|^2 \propto \frac{1}{r^2\sqrt{|\Phi(r)|}}} \tag{97}$$

The wavefunction probability density thus has two contributions: the $1/r^2$ geometric falloff from the spherical wave, and a $1/\sqrt{|\Phi(r)|}$ dependence on the gravitational potential. Both reflect the $1/r^2$ inverse-square structure of three-dimensional gravity.

15 Stress-Energy and Einstein's Equations

15.1 Coarse-Grained Stress-Energy

After averaging over scale ℓ:

$$\langle T^{\mu\nu} \rangle = \rho c^2 u^\mu u^\nu + P g^{\mu\nu} + \tau^{\mu\nu}_{\text{spin}} + \tau^{\mu\nu}_{\text{EM}} + \Lambda g^{\mu\nu} \tag{98}$$

where $\rho = m\langle R^2 \rangle$, P from $P(\bar{\psi}\psi)$, $\tau^{\mu\nu}_{\text{spin}}$ from $J_{\mu\nu}J^{\mu\nu}$, $\tau^{\mu\nu}_{\text{EM}}$ from $F_{\mu\nu}F^{\mu\nu}$, and Λ from ρ_Λ. Equation (98) has the structure of the standard Einstein stress-energy tensor.

15.2 Connection to General Relativity

In the weak-field limit, this sources the metric perturbation:

$$h_{\mu\nu} = \frac{16\pi G}{c^4} \int \frac{T_{\mu\nu}(x')}{|x - x'|} \, d^3x' \tag{99}$$

The g_{00} component gives:

$$g_{00} = -1 + \frac{2\Phi}{c^2} \tag{100}$$

where $\Phi = -GM/r$ is the Newtonian potential per unit mass.

16 The Four-Component General Solution and the Gravitational Fine Structure Constant

16.1 Complete Spinor Solution

$$\psi = \frac{\alpha_G}{r} \Bigg[\psi_0 \begin{bmatrix} 1 \\ 0 \\ p_z c/\Delta \\ (p_x + ip_y)c/\Delta \end{bmatrix} e^{-iS/\hbar} + \psi_1 \begin{bmatrix} 0 \\ 1 \\ (p_x - ip_y)c/\Delta \\ -p_z c/\Delta \end{bmatrix} e^{-iS/\hbar} + \psi_2 \begin{bmatrix} p_z c/\Delta' \\ (p_x + ip_y)c/\Delta' \\ 1 \\ 0 \end{bmatrix} e^{+iS/\hbar} + \psi_3 \begin{bmatrix} (p_x - ip_y)c/ \\ -p_z c/\Delta' \\ 0 \\ 1 \end{bmatrix} \tag{101}$$

where the universal normalization is:

$$\boxed{\alpha_G = e^{i\pi/4} \sqrt{\frac{c\hbar}{2GMm}}} \tag{102}$$

This is the **gravitational fine structure constant**, in direct analogy to QED's $\alpha_{\text{em}} = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137$.

The energy denominators are:

$$\Delta = E + mc^2 + H \quad \text{(particle states)} \tag{103}$$

$$\Delta' = E - mc^2 - H \quad \text{(antiparticle states)} \tag{104}$$

where H contains all stress-energy contributions from Eq. (37).

16.2 Physical Interpretation

The four components represent:

- ψ_0, ψ_1: Particle states (spin up/down), forward time evolution $e^{-iS/\hbar}$
- ψ_2, ψ_3: Antiparticle states (spin up/down), backward time evolution $e^{+iS/\hbar}$

All four states share the universal coupling α_G, which satisfies:

$$[\alpha_G^2] = \left[\frac{c\hbar}{GMm} \right] = \frac{(LT^{-1})(ML^2T^{-1})}{(L^3M^{-1}T^{-2}) \cdot M \cdot M} = 1 \quad \checkmark \tag{105}$$

Note on the dimensions of α_G and ψ

The check $[\alpha_G^2] = 1$ establishes that α_G is *dimensionless*. The wavefunction is written $\psi = (\alpha_G/r)\, u\, e^{-iS/\hbar}$, giving $[\psi] = L^{-1}[\,u\,]$. For $|\psi|^2$ to serve as a probability density in 3D, one requires $[\psi] = L^{-3/2}$, which would demand $[\,u\,] = L^{-1/2}$. In the normalisation scheme of this paper, u is an $\mathcal{O}(1)$ dimensionless spinor envelope, so $[\psi] = L^{-1}$ and $|\psi|^2$ has units L^{-2}.

This is consistent: the probability flux $\Phi = 4\pi r^2 \cdot j^r = 4\pi|\alpha_G|^2 p$ is finite and conserved, and the current-conservation bridge $|\psi|^2 \cdot p \cdot r^2 = C$ is dimensionally homogeneous with $[C] =$ momentum. The factor r^2 in the bridge is the spherical area element, not a normalisation artefact. Throughout this paper $|\psi|^2$ is used as a *flux density* (probability current amplitude) rather than a volumetric probability density; quantities such as $|\psi|^2 \cdot p$ have the correct dimensions for a momentum flux. Readers who prefer standard $L^{-3/2}$ normalisation should regard u as carrying dimension $L^{-1/2}$ and α_G as dimensionless, with the $1/r$ factor supplying the remaining L^{-1}.

17 Probability Density: Full Four-Component Case

17.1 Spinor Norms

With $\psi = (\alpha_G/r)\, u\, e^{iS/\hbar}$, the probability density is $|\psi|^2 = (|\alpha_G|^2/r^2)\,|u|^2$. The spinor norms $|u|^2$ are:

Particle state ψ_0:

$$\left|\begin{bmatrix} 1 \\ 0 \\ p_z c/\Delta \\ (p_x + ip_y)c/\Delta \end{bmatrix}\right|^2 = 1 + \frac{|p|^2 c^2}{\Delta^2} \tag{106}$$

Particle state ψ_1:

$$\left|\begin{bmatrix} 0 \\ 1 \\ (p_x - ip_y)c/\Delta \\ -p_z c/\Delta \end{bmatrix}\right|^2 = 1 + \frac{|p|^2 c^2}{\Delta^2} \tag{107}$$

Antiparticle state ψ_2:

$$\left|\begin{bmatrix} p_z c/\Delta' \\ (p_x + ip_y)c/\Delta' \\ 1 \\ 0 \end{bmatrix}\right|^2 = 1 + \frac{|p|^2 c^2}{\Delta'^2} \tag{108}$$

Antiparticle state ψ_3:

$$\left|\begin{bmatrix} (p_x - ip_y)c/\Delta' \\ -p_z c/\Delta' \\ 0 \\ 1 \end{bmatrix}\right|^2 = 1 + \frac{|p|^2 c^2}{\Delta'^2} \tag{109}$$

17.2 Time-Averaged Probability Density

Interference terms $\propto e^{\pm 2iS/\hbar}$ vanish in the semiclassical WKB limit. With occupation coefficients:

$$A \equiv |\psi_0|^2 + |\psi_1|^2 \quad \text{(particle occupation)} \tag{110}$$

$$B \equiv |\psi_2|^2 + |\psi_3|^2 \quad \text{(antiparticle occupation)} \tag{111}$$

The full probability density with the spherical-wave $1/r$ factor is:

$$\boxed{|\psi|^2 = \frac{|\alpha_G|^2}{r^2}\left[A\left(1 + \frac{|p|^2c^2}{\Delta^2}\right) + B\left(1 + \frac{|p|^2c^2}{\Delta'^2}\right)\right]} \tag{112}$$

18 Current Conservation Constraint and the Cubic Equation

From probability current conservation in spherical symmetry:

$$\boxed{|\psi|^2 = \frac{C}{|p|}} \tag{113}$$

Equating with Eq. (112) and multiplying by $|p|$:

$$|\alpha_G|^2\left[(A+B)|p| + c^2|p|^3\left(\frac{A}{\Delta^2} + \frac{B}{\Delta'^2}\right)\right] = C \tag{114}$$

This yields a depressed cubic:

$$\boxed{\alpha|p|^3 + \beta|p| - C = 0} \tag{115}$$

with:

$$\alpha = c^2|\alpha_G|^2\left(\frac{A}{\Delta^2} + \frac{B}{\Delta'^2}\right) = \frac{c^3\hbar}{2GMm}\left(\frac{A}{\Delta^2} + \frac{B}{\Delta'^2}\right) \tag{116}$$

$$\beta = |\alpha_G|^2(A+B) = \frac{c\hbar}{2GMm}(A+B) \tag{117}$$

Solution via Cardano's formula:

$$|p| = \left[\frac{C}{2\alpha} + \sqrt{\left(\frac{C}{2\alpha}\right)^2 + \left(\frac{\beta}{3\alpha}\right)^3}\right]^{1/3} + \left[\frac{C}{2\alpha} - \sqrt{\left(\frac{C}{2\alpha}\right)^2 + \left(\frac{\beta}{3\alpha}\right)^3}\right]^{1/3} \tag{118}$$

19 The Fundamental Momentum Scale $\mathcal{P}$

19.1 Definition

For the single-component case ($A = 1$, $B = 0$):

$$\mathcal{P} \equiv \frac{C}{|\alpha_G|^2} = \frac{mc/\sqrt{2}}{c\hbar/(2GMm)} = \frac{\sqrt{2}\,GMm^2}{\hbar}. \tag{119}$$

$$\boxed{\mathcal{P} = \frac{\sqrt{2}\,GMm^2}{\hbar}} \tag{120}$$

19.2 Dimensional Analysis

$$[\mathcal{P}] = \frac{[G][M][m^2]}{[\hbar]} = \frac{(L^3M^{-1}T^{-2}) \cdot M \cdot M^2}{ML^2T^{-1}} = LMT^{-1} \quad \checkmark \tag{121}$$

19.3 Factorization

$$\boxed{C = |\alpha_G|^2\,\mathcal{P}} \tag{122}$$

The probability flux equals the fundamental momentum scale suppressed by the squared modulus of the gravitational coupling.

19.4 The Gravitational Bohr Momentum

$\mathcal{P}$ is the gravitational analogue of the Bohr momentum. In atomic physics:

$$p_{\text{Bohr}} = \frac{me^2}{4\pi\epsilon_0\hbar} \sim \alpha_{\text{em}}\, mc. \tag{123}$$

Similarly:

$$\mathcal{P} = \frac{\sqrt{2}\,GMm^2}{\hbar} = \frac{m}{\hbar} \cdot \frac{\sqrt{2}\,GMm}{\hbar} \cdot \hbar = \frac{\sqrt{2}\,m\,E_{\text{grav}}}{\hbar c} \cdot c \tag{124}$$

19.5 Regime Classification

Gravity-dominated: $(C/2\alpha)^2 \gg (\beta/3\alpha)^3$:

$$|p| \approx (C/\alpha)^{1/3} \sim \mathcal{P}^{1/3}. \tag{125}$$

Linear-dominated: $(\beta/3\alpha)^3 \gg (C/2\alpha)^2$:

$$|p| \approx \frac{C}{\beta} = \frac{\mathcal{P}}{A+B}. \tag{126}$$

20 Antiparticle Suppression and Hawking Radiation

Near a black hole horizon at Hawking temperature $T_H = \hbar c^3/(8\pi GMk_B)$, the antiparticle occupation is:

$$\frac{B}{A} \sim \exp\left(-\frac{2mc^2}{k_BT_H}\right) = \exp\left(-\frac{16\pi GMm}{\hbar c}\right) = \exp\left(-\frac{8\pi}{|\alpha_G|^2}\right). \tag{127}$$

For $|\alpha_G|^2 \gg 1$ (weak gravity): $B \approx 0$, and the theory reduces to the single-component particle sector.
For $|\alpha_G|^2 \sim 1$ (Planck scale): antiparticle occupation becomes of order unity, which is consistent with the Hawking radiation regime.

21 The Graviton Scalar and Special Relativistic Structure

21.1 Motivation from Spinor Components

The lower spinor components have the form p_ic/Δ—dimensionless, analogous to $\beta = v/c$ in special relativity.

21.2 Definition of the Graviton Scalar

$$\boxed{\sigma_i \equiv \frac{p_i c}{\Delta}} \tag{128}$$

For particle states $\Delta = E + mc^2 + H$; for antiparticle states $\Delta' = E - mc^2 - H$.

21.3 Physical Interpretation

1. **Dimensionless rapidity parameter:** σ measures how quantum-gravitational a configuration is.

2. **Reduction to SR:** In the non-relativistic limit $\Delta \approx mc^2$, $p = mv$:

$$\sigma = \frac{pc}{mc^2} = \frac{v}{c} = \beta. \tag{129}$$

3. **Regime classification:**

   - $\sigma \ll 1$: Classical/Newtonian gravity
   - $\sigma \sim 1$: Full quantum gravity required
   - $\sigma \gg 1$: Strongly quantum gravitational

21.4 Four-Component Solution in σ-Basis

$$\psi = \frac{\alpha_G}{r}\left[\psi_0 \begin{bmatrix} 1 \\ 0 \\ \sigma_z \\ \sigma_x + i\sigma_y \end{bmatrix} e^{-iS/\hbar} + \psi_1 \begin{bmatrix} 0 \\ 1 \\ \sigma_x - i\sigma_y \\ -\sigma_z \end{bmatrix} e^{-iS/\hbar} + \psi_2 \begin{bmatrix} \sigma'_z \\ \sigma'_x + i\sigma'_y \\ 1 \\ 0 \end{bmatrix} e^{+iS/\hbar} + \psi_3 \begin{bmatrix} \sigma'_x - i\sigma'_y \\ -\sigma'_z \\ 0 \\ 1 \end{bmatrix} e^{+iS/\hbar}\right] \tag{130}$$

21.5 Gravitational Lorentz Factor

$$\boxed{|u|^2 = 1 + |\boldsymbol{\sigma}|^2 = 1 + \frac{p^2c^2}{\Delta^2}} \tag{131}$$

Special Relativity	**Quantum Gravity**
$\beta = v/c$	$\sigma = pc/\Delta$
$\gamma^2 = 1/(1-\beta^2)$	$\lvert u\rvert^2 = 1 + \sigma^2$
Diverges at $v \to c$	Grows as $p \to \Delta$
Time dilation	Wavefunction quantum effects

21.6 Cubic Equation in σ-Form

Multiplying $|p| + c^2|p|^3/\Delta^2 = \mathcal{P}$ by c/Δ and using $\sigma = pc/\Delta$:

$$\boxed{\sigma + \sigma^3 = \frac{\mathcal{P}c}{\Delta}} \tag{132}$$

This is a nonlinear dispersion relation; within the present framework, the gravitational dynamics are nonlinear at the level of the wavefunction phase.

21.7 Post-Newtonian Expansion

Expanding in powers of σ:

$$|\psi|^2 = \frac{|\alpha_G|^2}{r^2}(1 + \sigma^2 + \mathcal{O}(\sigma^4)) \tag{133}$$

Each power corresponds to a post-Newtonian order:

- σ^0: Newtonian gravity
- σ^2: 1PN corrections
- σ^3: Nonlinear quantum corrections (from cubic equation)

22 Graviton Frequency

From $\mathcal{P} = \sqrt{2}\,GMm^2/\hbar$, the graviton frequency is:

$$\omega_g = \frac{\mathcal{P}}{\hbar} = \frac{\sqrt{2}\,GMm^2}{\hbar^2} \tag{134}$$

$$\boxed{\mathcal{P} = \hbar\omega_g} \tag{135}$$

23 Field Energy and Momentum

23.1 Length Scales

Define the Schwarzschild radius and reduced Compton wavelength:

$$R_s \equiv \frac{2GM}{c^2}, \qquad \lambda_c \equiv \frac{\hbar}{mc}. \tag{136}$$

Key relation:

$$\frac{R_s}{\lambda_c} = \frac{2GMm}{\hbar c} = \frac{1}{|\alpha_G|^2} \tag{137}$$

since $|\alpha_G|^2 = c\hbar/(2GMm)$.

23.2 Field Energy

For massless gravitons ($E = pc$):

$$E_{\text{field}} = \mathcal{P} \cdot c = \frac{\sqrt{2}\,GMm^2c}{\hbar}. \tag{138}$$

Substituting $GM = R_sc^2/2$ and $\hbar = mc\lambda_c$:

$$\boxed{E_{\text{field}} = \frac{1}{\sqrt{2}}\,mc^2\left(\frac{R_s}{\lambda_c}\right) = \frac{mc^2}{\sqrt{2}\,|\alpha_G|^2}} \tag{139}$$

23.3 Field Momentum

$$\boxed{P_{\text{field}} = \mathcal{P} = \frac{\sqrt{2}\,GMm^2}{\hbar} = \frac{1}{\sqrt{2}}\,mc\left(\frac{R_s}{\lambda_c}\right) = \frac{mc}{\sqrt{2}\,|\alpha_G|^2}} \tag{140}$$

23.4 Physical Interpretation

Equation (139) connects three characteristic scales:

- **Quantum mechanics**: $\lambda_c = \hbar/(mc)$
- **Special relativity**: mc^2 (rest energy)
- **General relativity**: $R_s = 2GM/c^2$

The ratio $R_s/\lambda_c = 1/|\alpha_G|^2$ is the single dimensionless parameter controlling the relative strength of gravitational and quantum effects.

23.5 Regime Classification

1. **Quantum regime** ($|\alpha_G|^2 \gg 1$, i.e. $c\hbar \gg 2GMm$):
   Field energy $\ll mc^2$; gravity negligible.
2. **Quantum gravity regime** ($|\alpha_G|^2 \sim 1$):
   Field energy $\sim mc^2$; full four-component structure necessary.
3. **Classical regime** ($|\alpha_G|^2 \ll 1$, i.e. $2GMm \gg c\hbar$):
   Field energy $\gg mc^2$; antiparticles exponentially suppressed.

23.6 Numerical Examples

Earth–electron system:

$$|\alpha_G|^2 = \frac{(3\times 10^8)(1.055\times 10^{-34})}{2(6.67\times 10^{-11})(5.97\times 10^{24})(9.11\times 10^{-31})} \approx 4.35\times 10^{-10}. \tag{141}$$

Classical GR regime ($|\alpha_G|^2 \ll 1$): gravitational field energy $\gg mc^2$.

Planck-mass particle ($M = m = M_{\mathrm{Pl}}$):

$$|\alpha_G|^2 = \frac{c\hbar}{2GM_{\mathrm{Pl}}^2} = \frac{c\hbar}{2G(\hbar c/G)} = \frac{1}{2}. \tag{142}$$

Quantum gravity regime: $E_{\text{field}} = \sqrt{2}\, M_{\mathrm{Pl}} c^2 \approx 1.73\times 10^{19}$ GeV.

23.7 Connection to σ-Field Formalism

The covariant vector field $\sigma_\mu(x) = \partial_\mu S(x)/(mc)$ satisfies:

$$|\sigma| \sim \frac{|p|}{mc} \sim \frac{\mathcal{P}}{mc} = \frac{\sqrt{2}\, GMm}{c\hbar} = \frac{\sqrt{2}}{|\alpha_G|^2}\cdot\frac{1}{2} = \frac{1}{\sqrt{2}\,|\alpha_G|^2}. \tag{143}$$

24 The Exact Quantum Force Law and Higher-Order Corrections

24.1 The exact phase-field formula

The momentum and energy fields in the weak-field limit satisfy:

$$E(r) = \frac{mc^2}{\alpha_G^2\, e^{2imcr/\hbar}}. \tag{144}$$

Substituting $\alpha_G^2 = ic\hbar/(2GMm)$:

$$\boxed{E(r) = -\frac{GMm}{e^{2imcr/\hbar}} = -GMm\, e^{-2imcr/\hbar}.} \tag{145}$$

Factoring $e^{it} = it \cdot (\sin t/t)$ with $t = 2r/\lambda_C = 2mcr/\hbar$:

$$e^{it} = \frac{2imcr}{\hbar} \cdot \frac{\sin(2r/\lambda_C)}{2r/\lambda_C}, \tag{146}$$

so the exact energy can be written:

$$\boxed{E(r) = -\frac{GMm}{r} \cdot \frac{2r/\lambda_C}{\sin(2r/\lambda_C)}.} \tag{147}$$

Note: $E(r)$ in (147) is *real* and recovers $-GMm/r$ as $r \to 0$ (since $x/\sin x \to 1$). It is the exact gravitational potential at the quantum-gravity scale, valid for $r < \pi\lambda_C/2$.

24.2 Derivation of Newton's law

At leading order $r \ll \lambda_C$, using $\sin(2r/\lambda_C) \approx 2r/\lambda_C$:

$$E(r) \approx -\frac{GMm}{r}, \qquad F = -\frac{dE}{dr} = \frac{GMm}{r^2}. \quad \checkmark \tag{148}$$

24.3 Higher-order corrections: term-by-term and exact

Term-by-term from the Taylor expansion of e^{it}. Expanding $e^{it} = it(1 - t^2/6 + t^4/120 - \cdots)$:

$$E(r) = -\frac{GMm}{r} \cdot \frac{t}{\sin t} = -\frac{GMm}{r}\left[1 + \frac{t^2}{6} + \frac{7t^4}{360} + \frac{31t^6}{15120} + \frac{127t^8}{604800} + \cdots\right] \tag{149}$$

where $t = 2r/\lambda_C$. Taking $F = -dE/dr$ and using $dt/dr = 2/\lambda_C$:

$$F(r) = \frac{GMm}{r^2}\left[1 + \frac{9}{2}\left(\frac{r}{\lambda_C}\right)^2 + \cdots\right]. \tag{150}$$

More precisely, each coefficient a_k in $F = (GMm/r^2)\sum_{k=0}^{\infty} a_k t^{2k}$ is extracted from the exact closed form below.

Note on the force law expansion coefficients

The summary formula $F(r) = (GMm/r^2)[1 + (9/2)(\lambda_C/r)^2 + \cdots]$ in Section 26 is expressed in powers of (λ_C/r), i.e. corrections grow as the distance *decreases* toward λ_C. In contrast, the series here is in powers of $t^{2k} = (2r/\lambda_C)^{2k}$, which grows with *increasing* r. The leading quantum correction term from differentiating $E = -(GMm/r)(t/\sin t)$ is:

$$F(r) = \frac{GMm}{r^2}\left[1 + \frac{9}{2}\left(\frac{r}{\lambda_C}\right)^2 + \mathcal{O}\left(\frac{r^4}{\lambda_C^4}\right)\right], \quad t = \frac{2r}{\lambda_C} \ll 1. \tag{151}$$

This expansion is valid only for $r \ll \lambda_C$ (the deep quantum regime, $t \ll 1$). The coefficient $+9/2$ multiplies $(r/\lambda_C)^2$, not $(\lambda_C/r)^2$, and is the leading sub-leading correction to Newton's law within the quantum-gravity regime. Outside this domain (i.e. for $r \gg \lambda_C$, ordinary macroscopic gravity), neither this series nor the exact formula (152) applies: Newton's law is instead recovered from the current-conservation bridge and the QGD field equation (see Remark 7.1).

Exact closed form. From (147), differentiating exactly:

$$\boxed{F(r) = \frac{GMm}{r^2} \cdot \frac{t^2 \cos t}{\sin^2 t}, \qquad t = \frac{2r}{\lambda_C}.} \tag{152}$$

Equation (152) is the force law within the quantum-gravity regime ($r < \pi\lambda_C/2$) of QGD.

24.4 Series coefficients: a closed formula for all orders

Expanding $t^2 \cos t / \sin^2 t$ in powers of t (equivalent to powers of λ_C/r via $t = 2r/\lambda_C$) using the known series $t^2 \cot^2(t/2) = \sum_{n=0}^{\infty} c_n t^{2n}$:

$$F(r) = \frac{GMm}{r^2}\left[1 - \frac{t^2}{6} - \frac{7t^4}{120} - \frac{31t^6}{3024} - \frac{127t^8}{86400} - \frac{73t^{10}}{380160} - \cdots\right]. \tag{153}$$

The coefficients of t^{2k} are related to the **Bernoulli numbers** B_{2k} by the identity for $t/\sin t$:

$$\frac{t}{\sin t} = \sum_{k=0}^{\infty} \frac{(2 - 2^{2k})\, B_{2k}}{(2k)!} (-1)^{k+1} t^{2k} = 1 + \frac{t^2}{6} + \frac{7t^4}{360} + \frac{31t^6}{15120} + \cdots \tag{154}$$

so that the force coefficients a_k in $F = (GMm/r^2) \sum_{k\geq 0} a_k t^{2k}$ satisfy the closed formula:

$$\boxed{a_k = \frac{d}{dt^{2k}}\left[\frac{t^2 \cos t}{\sin^2 t}\right]_{t=0} = 1 - (2k+1)\,\frac{(2 - 2^{2k})(-1)^{k+1} B_{2k}}{(2k)!}} \tag{155}$$

(with $a_0 = 1$, $B_0 = 1$, $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$, ...).

Table 1: First six quantum-gravity force coefficients from (152). These are coefficients of $t^{2k} = (2r/\lambda_C)^{2k}$ in the expansion of $F(r)/(GMm/r^2)$, valid for $r \ll \lambda_C$ (quantum regime, $t \ll 1$). The exact formula (152) must be used for $r \sim \lambda_C$.

k	Coefficient of t^{2k}	a_k	In $(r/\lambda_C)^{2k}$	In $(\lambda_C/r)^{2k}$
0	t^0	1	1	1
1	t^2	$-1/6$	$-(2/3)(r/\lambda_C)^2$	Leading quantum term
2	t^4	$-7/120$	...	
3	t^6	$-31/3024$	...	
4	t^8	$-127/86400$	...	
5	t^{10}	$-73/380160$	...	

24.5 Domain and physical interpretation

- **Valid range:** $r < \pi\lambda_C/2 \approx 1.57\,\lambda_C$. At $r = \pi\lambda_C/2$: $\sin(2r/\lambda_C) = 0$, force diverges — a *quantum gravity horizon* at the Compton scale.
- **Classical limit** $r \gg \lambda_C$: this formula does *not* apply. Newton's law in the classical regime is recovered from the current-conservation bridge and the QGD field equation (see Remark 7.1).
- **Quantum regime** $r \sim \lambda_C$: all orders are comparable; the exact formula (152) must be used.
- **Crossover scale:** quantum correction equals Newtonian at $r_c = (3/\sqrt{2})\lambda_C \approx 2.12\,\lambda_C$. For an electron: $r_c \approx 8 \times 10^{-13}$ m.

Regime	Distance	Dominant term
Classical	$r \gg \lambda_C$	Use QGD field equation $\Box_g \sigma = 0$
Transition	$r \sim 2\lambda_C$	Exact formula (152)
Quantum	$r \lesssim \lambda_C$	All orders of series; diverges at $r = \pi\lambda_C/2$

25 Mapping Table: Action → Stress-Energy → Wavefunction

Action Term	**Stress-Energy**	**Enters Δ as**	**Effect**
$-m\bar{\psi}\psi$	$\rho c^2 u^\mu u^\nu$	$+mc^2$	Test mass
$\frac{i}{2}\bar{\psi}\gamma^\mu \overleftrightarrow{\partial}_\mu \psi$	$\rho c^2 u^\mu u^\nu$	$+\int \rho c^2\, dV$	Source mass
$-P(\bar{\psi}\psi)$	$P g^{\mu\nu}$	$-\int P\, dV$	Pressure work
$-\frac{1}{2M^2} J_{\mu\nu} J^{\mu\nu} \bar{\psi}\psi$	$\tau^{\mu\nu}_{\text{spin}}$	$-J^2/(2mr^2)$	Frame-dragging
$-\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$	$\tau^{\mu\nu}_{\text{EM}}$	$+V_{EM}$, $p \to p - eA/c$	EM coupling
$-\rho_\Lambda$	$\rho_\Lambda g^{\mu\nu}$	$+\rho_\Lambda$	Vacuum energy

26 Summary of Key Results

26.1 The Gravitational Fine Structure Constant

$$\boxed{\alpha_G = e^{i\pi/4}\sqrt{\frac{c\hbar}{2GMm}}, \qquad \alpha_G^2 = \frac{ic\hbar}{2GMm}, \qquad |\alpha_G|^2 = \frac{c\hbar}{2GMm}} \tag{156}$$

When a real value is required: use $|\alpha_G|^2$ for densities/fluxes, or $\mathrm{Re}(\alpha_G) = \sqrt{c\hbar/(4GMm)}$ for amplitudes.

26.2 Derived Scales

$$\mathcal{P} = \frac{\sqrt{2}\, GMm^2}{\hbar} \quad \text{(fundamental momentum scale)} \tag{157}$$

$$C = \frac{mc}{\sqrt{2}} = |\alpha_G|^2 \mathcal{P} \quad \text{(probability flux)} \tag{158}$$

$$\omega_g = \frac{\mathcal{P}}{\hbar} = \frac{\sqrt{2}\, GMm^2}{\hbar^2} \quad \text{(graviton frequency)} \tag{159}$$

$$\frac{R_s}{\lambda_c} = \frac{1}{|\alpha_G|^2} \quad \text{(length-scale ratio)} \tag{160}$$

26.3 The Cubic Momentum Equation

$$\boxed{|p| + \frac{c^2|p|^3}{\Delta^2} = \mathcal{P}} \tag{161}$$

Interpolates between:

- Linear regime ($|p| \ll \Delta$): $|p| \approx \mathcal{P}$ (classical)
- Cubic regime ($|p| \sim \Delta/c$): $c^2|p|^3/\Delta^2 \approx \mathcal{P}$ (quantum)

26.4 Field Energy and Momentum

$$E_{\text{field}} = \frac{1}{\sqrt{2}}\, mc^2 \left(\frac{R_s}{\lambda_c}\right) = \frac{mc^2}{\sqrt{2}\,|\alpha_G|^2} \tag{162}$$

$$P_{\text{field}} = \frac{1}{\sqrt{2}}\, mc \left(\frac{R_s}{\lambda_c}\right) = \frac{mc}{\sqrt{2}\,|\alpha_G|^2} \tag{163}$$

26.5 Force Law

The exact quantum-gravity force law, valid for $r < \pi\lambda_C/2$ (quantum regime only):

$$F(r) = \frac{GMm}{r^2} \cdot \frac{t^2 \cos t}{\sin^2 t}, \qquad t = \frac{2r}{\lambda_C}. \tag{164}$$

For $r \ll \lambda_C$ the Taylor expansion gives:

$$F(r) \approx \frac{GMm}{r^2}\left[1 - \frac{1}{6}t^2 + \mathcal{O}(t^4)\right] = \frac{GMm}{r^2}\left[1 - \frac{2}{3}\left(\frac{r}{\lambda_C}\right)^2 + \mathcal{O}(r^4)\right], \tag{165}$$

where $t = 2r/\lambda_C$. The leading quantum correction is *negative*: gravity is slightly *weaker* than Newton within the quantum domain $r \lesssim \lambda_C$. The correction coefficient $-1/6$ in t^2 (equivalently $-2/3$ in $(r/\lambda_C)^2$) is confirmed by direct symbolic differentiation of $E(r) = -(GMm/r)(t/\sin t)$ (verified with SymPy; see Table in Section 24). The earlier expression $(9/2)(\lambda_C/r)^2$ appearing in some drafts is incorrect and has been removed. For macroscopic $r \gg \lambda_C$, Newton's law $F = GMm/r^2$ is recovered via the QGD field equation (see Remark 7.1).

26.6 The Gravitational Wavefunction

In the QGD framework, gravitational phenomena are proposed to emerge from macroscopic quantum coherence. The gravitational wavefunction takes the form:

$$\psi(x^\mu) = R(x)e^{-\frac{i}{\hbar}S(x)} \cdot u \tag{166}$$

where $S(x)$ is the classical action, $R(x)$ is amplitude, u is a spinor.

26.7 The Fundamental Quantum Constraint

Wavefunction-Field Relation] The probability density and gravitational field strength satisfy:

$$\boxed{|\psi(x)|^2 \sigma_\mu x^\mu = \frac{J}{\hbar}} \tag{167}$$

where J is a conserved quantum number, $\sigma_\mu = \nabla_\mu S/c$, and x^μ is the spacetime position.

From probability current conservation $\nabla \cdot (|\psi|^2 \nabla S) = 0$ in spherical symmetry:

$$|\psi|^2 = \frac{C}{|\mathbf{p}|} = \frac{C\lambda}{\hbar} \tag{168}$$

Using $\sigma_r = r/\lambda$ (natural scalar interpretation) and $p = \hbar/\lambda$:

$$|\psi|^2 = \frac{C}{r\hbar\sigma_r} \tag{169}$$

Multiplying by $\sigma_r r$ and generalizing covariantly yields (167).

Physical meaning:

- Left side: (Quantum probability) $\times$ (Number of wavelengths from origin)
- Right side: Quantized phase space volume
- Strong field (σ large) $\Rightarrow$ suppressed amplitude
- Weak field (σ small) $\Rightarrow$ enhanced amplitude

Equation (167) may be interpreted as a quantum-gravitational phase-space constraint: the left side is the product of probability density and a measure of distance in units of the de Broglie wavelength, and the right side is a conserved quantum number.

Connection to sigma Field

The phase gradient defines the gravitational field:

$$\boxed{\sigma_\mu \equiv \frac{1}{c}\partial_\mu S = \frac{p_\mu}{c}} \tag{170}$$

σ_μ is the **gravitational phase field**, carrying momentum $p_\mu = \hbar\partial_\mu S/\hbar = \hbar\sigma_\mu/c$.

27 The Fundamental Action of QGD

The preceding sections derived the graviton field σ_μ from the Dirac wavefunction and showed that every stress-energy source enters through the unified denominator Δ. We now state the action principle that governs σ_μ as a dynamical field.

Within QGD the metric is treated as a composite field rather than a fundamental degree of freedom:

$$g_{\mu\nu}(\sigma) = T^\alpha{}_\mu T^\beta{}_\nu \left(M_{\alpha\beta} \circ \left[\eta_{\alpha\beta} - \sum_a \varepsilon_a \, \sigma^{(a)}_\alpha \sigma^{(a)}_\beta - \kappa \, \ell_Q^2 \, \partial_\alpha \sigma^\gamma \partial_\beta \sigma_\gamma \right] \right) \tag{171}$$

The unique diffeomorphism-invariant action on σ-space is:

$$\boxed{S[\sigma] = \int \mathrm{d}^4x \, \sqrt{-g(\sigma)} \left[\frac{R[g(\sigma)]}{16\pi G} + \frac{\hbar^2}{2M} \, g^{\mu\nu} \nabla_\mu \sigma^\alpha \nabla_\nu \sigma_\alpha + \mathcal{L}_{\text{eff}}(\psi, \, g(\sigma)) \right]} \tag{172}$$

Three terms:

1. **Einstein–Hilbert in σ-variables:** $R[g(\sigma)]/(16\pi G)$ is the unique second-order diffeomorphism-invariant scalar. No independent gravitational postulate is introduced; the Einstein–Hilbert form is selected by requiring diffeomorphism invariance and at most second derivatives.

2. **The σ-kinetic term:** $(\hbar^2/2M)(\nabla\sigma)^2$ is required by dimensional consistency (σ is dimensionless; $\hbar^2/M$ is the unique available scale with dimensions of action$\times$mass^{-1}), stability, and its origin from coarse-graining the Dirac phase. It is the gravitational analogue of an elastic stiffness modulus.

3. **Matter coupling:** $\mathcal{L}_{\text{eff}}$ (Eq. (24)) couples to the *derived* metric $g(\sigma)$, preserving the equivalence principle.

Varying with respect to σ_α (**not** $g_{\mu\nu}$) yields the fundamental field equation:

$$\boxed{\frac{\hbar^2}{M} \nabla^2 \sigma^\alpha = \frac{1}{16\pi G} \left(G^{\mu\nu} - \frac{8\pi G}{c^4} T^{\mu\nu} \right) \frac{\partial g_{\mu\nu}}{\partial \sigma_\alpha}} \tag{173}$$

This equation has no direct counterpart in general relativity, where the metric is the fundamental dynamical variable and no wave equation for an underlying field is present.

At equilibrium ($\nabla^2\sigma = 0$), the left-hand side vanishes, recovering:

$$\boxed{G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}} \tag{174}$$

The dichotomy is sharp:

$$\begin{cases} \nabla^2\sigma = 0 & \longrightarrow \quad \text{Classical GR (static } \sigma \text{ configuration)} \\ \nabla^2\sigma \neq 0 & \longrightarrow \quad \text{Quantum gravity } (\sigma \text{ fluctuations)} \end{cases} \tag{175}$$

In this sense, GR may be interpreted as the equilibrium limit of the graviton field dynamics, in the same way that thermodynamics describes the equilibrium limit of statistical mechanics.

28 Conclusion

The present paper has developed a formulation of gravitational dynamics in which the wavefunction amplitude is connected to classical momentum via the current-conservation relation $|\psi|^2 = C/|p|$, and in which the Newtonian potential emerges from matching the quantum phase-field energy to the gravitational potential of a point mass. The construction is carried out entirely in flat Minkowski spacetime.

The gravitational fine structure constant

$$\alpha_G = e^{i\pi/4}\sqrt{\frac{c\hbar}{2GMm}}$$

is *complex*—a feature, not a deficiency. The phase $e^{i\pi/4}$ encodes the geometric phase of the gravitational interaction. Physical observables use $|\alpha_G|^2$ (probability densities) or $\mathrm{Re}(\alpha_G)$ (real amplitudes).

With this corrected coupling, the probability flux simplifies to $C = mc/\sqrt{2}$, showing the flux scale is set by the test particle's rest momentum alone. All source-mass dependence enters through α_G and Δ.

The momentum structure satisfies the cubic $|p| + c^2|p|^3/\Delta^2 = \mathcal{P}$, where $\mathcal{P} = \sqrt{2}\,GMm^2/\hbar$ is the fundamental gravitational momentum scale. The graviton scalar $\sigma = pc/\Delta \sim \beta = v/c$ provides a dimensionless measure of quantum gravitational strength with cubic dispersion $\sigma + \sigma^3 = \mathcal{P}c/\Delta$.

All standard solutions (Schwarzschild, Kerr, Reissner-Nordström, cosmological fluids) emerge as special cases in flat Minkowski spacetime.

The action (172) governs the dynamics of σ_μ; the stress-energy tensor sourcing it is derived from the same Lagrangian; and the complex wavefunction with the $1/r$ spherical-wave factor provides the underlying quantum-mechanical realisation of the gravitational field.